

Name: Solutions

Teacher/Class: _____

SYDNEY TECHNICAL HIGH SCHOOL



HSC ASSESSMENT TASK 1

DECEMBER 2009

MATHEMATICS

Time Allowed: 70 minutes

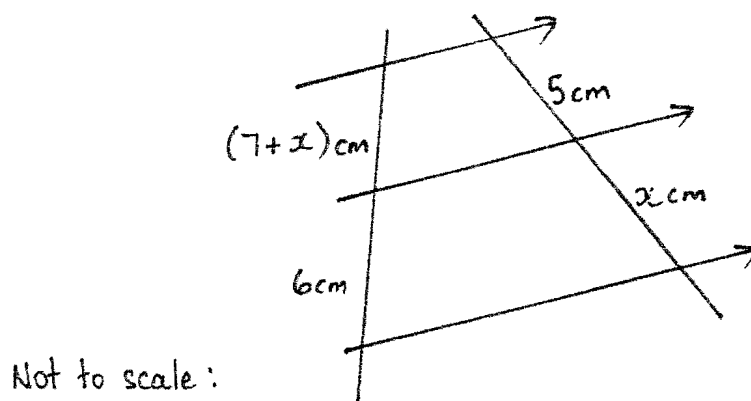
Instructions:

- Write your name and class at the top of each page
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start **each** question on a **new** page.
- Diagrams unless otherwise stated are **not** to scale

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	TOTAL
/8	8	/8	/8	/8	/8	/8	/56

Question 1 (8 marks)

- a) i) Solve $2x^2 - x = 1$ (3)
ii) Hence, solve $2x^2 - x - 1 > 0$
- b) What is the vertex of $y = 3x^2 - 2x + 1$? (2)
- c) What is the focal length of $x^2 = -12y$? (1)
- d) Find the value of x in : (2)

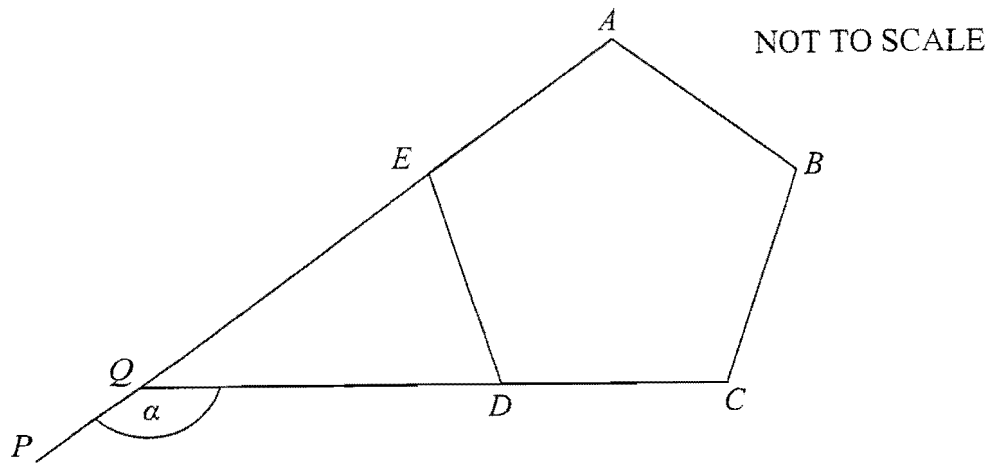


Question 2 (8 marks) **Start a new page**

- a) The roots of $5x^2 - 6x - 3 = 0$ are α and β (3)
- i) Find the value of,
(I) $\alpha + \beta$
(II) $\alpha^2 \beta + \alpha \beta^2$
- ii) Write down a quadratic equation, in the form $ax^2 + bx + c = 0$, where (3)
 a, b and c are integers, whose roots are 2α and 2β .
- b) Find the locus of a point P which moves so that it is always 4 units from the (2)
point $(-1, 2)$

Question 3 (8 marks) **Start a new page**

- a) ABCDE is a regular pentagon. The points P, Q, E and A are collinear.
CD is produced to meet PA at Q.



Find the size of angle α giving reasons. (2)

- b) Consider the parabola $x^2 = 8y + 16$
- i) Find the co-ordinates of the vertex (2)
 - ii) Find the co-ordinates of the focus (2)
 - iii) Find the equation of the tangent to the parabola at the point $(2, -\frac{3}{2})$ (2)

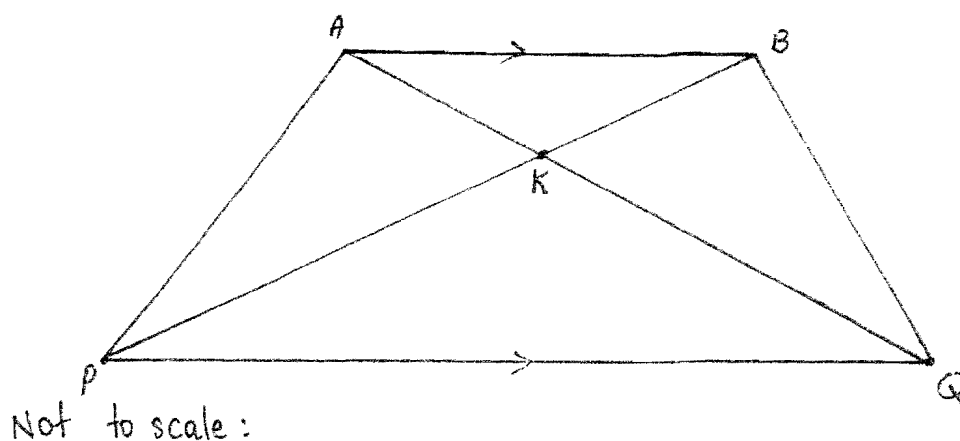
Question 4 (8 marks) [Start a new page](#)

- a) Find the radius and the centre of the circle whose equation is:

$$x^2 - 4x + y^2 + 6y - 12 = 0$$

(2)

- b) In the diagram $AB \parallel PQ$,
AQ bisects PB at K and
 $PK = KQ$



Copy the diagram onto your answer page, marking on it all relevant information.

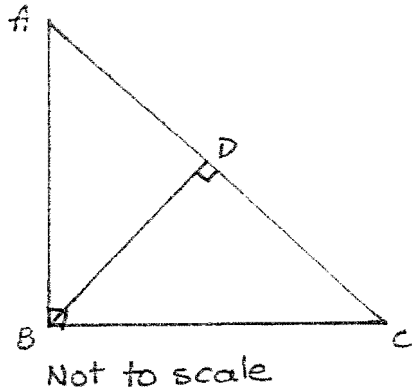
- i) Prove that $\triangle AKB$ is isosceles (3)
- ii) Prove that $\triangle AKP \equiv \triangle BKQ$ (2)
- iii) Hence, show that $AP = BQ$ (1)

Question 5 (8 marks) [Start a new page](#)

- a) Solve, $4^x - 9(2^x) + 8 = 0$ (3)
- b) Find the value of A, B and C, given
 $3x^2 + 4 \equiv A(x+2)^2 + B(x+2) + C$ (3)
- c) Sketch the parabola $y^2 = 12x$ showing its focus and directrix (2)

Question 6 (8 marks) **Start a new page**

- a) ABC is a triangle in which $\angle ABC = 90^\circ$ and $BD \perp AC$



- i) Show that $\triangle ABD$ is similar to $\triangle ACB$ (2)

- ii) If $AB = 3\text{cm}$ and $BC = 4\text{cm}$, find the length of BD , with reasons (2)

- b) Consider the quadratic expression $3kx^2 - 5x + 3k$

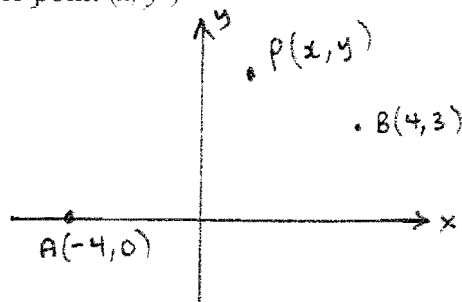
- i) Write down an expression for the discriminant of this quadratic. (1)

- ii) Hence, find the value/s of k for which $3kx^2 - 5x + 3k$ is negative definite. (3)

Question 7 (8 marks) **Start a new page**

- a) Let A and B be the fixed points $(-4, 0)$ and $(4, 3)$ respectively.

P is the variable point (x, y)



- i) Write down expressions for PA^2 and PB^2 in terms of x and y (2)

- ii) Hence, or otherwise, find the locus of the point P , such that, $PA = 2PB$ (3)

- b) Find the value/s of k for which the quadratic equation, $(k - 3)x^2 - 3kx + 25 = 0$ has one root double the other root. (3)

End of paper

Answers

Question 1

a) $2x^2 - x - 1 = 0$
 $(2x+1)(x-1) = 0$
 $x = -1/2, x = 1$

ii)  $x < -1/2, x > 1$

b) $x = \frac{2}{6}$ Vertex $(\frac{1}{3}, \frac{2}{3})$

c) $x^2 = -12y$
 - focal length 3 units

d) $\frac{x+7}{6} = \frac{5}{x}$
 $x^2 + 7x = 30 \quad x > 0$
 $(x+10)(x-3) = 0$
 $\therefore x = 3$

Question 2.

a) $\alpha + \beta = -\frac{b}{a}$
 $= \frac{6}{5}$

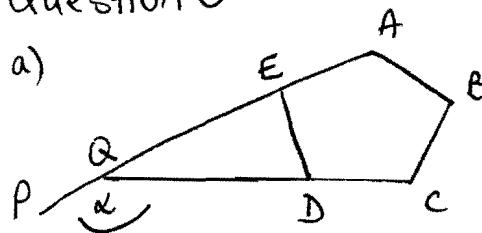
ii) $\alpha\beta(\alpha + \beta) = \frac{c}{a} \times \frac{6}{5}$
 $= \frac{-3}{5} \times \frac{6}{5}$
 $= \frac{-18}{25}$

ii. $x^2 - (2\alpha + 2\beta)x + 2\alpha \cdot 2\beta = 0$
 $x^2 - 2(\alpha + \beta)x + 4\alpha\beta = 0$
 $x^2 - 2(\frac{6}{5})x + 4(\frac{-3}{5}) = 0$
 $5x^2 - 12x - 12 = 0$

b) $\sqrt{(x+1)^2 + (y-2)^2} = 4$

$\therefore (x+1)^2 + (y-2)^2 = 16.$

Question 3



$\angle AED = \angle CDE = 108^\circ$

(angles of a regular pentagon)

$\angle QED = \angle EDQ = 72^\circ$

(straight line equals 180°)

$\alpha = \angle QED + \angle EDQ$ (exterior $\angle$ of $\triangle QED$)
 $\alpha = 144^\circ$

b) $x^2 = 8(y+2)$ ② $a = 2$

i. $V(0, -2)$ ②

ii. $S(0, 0)$ ②

iii. $8y = x^2 - 16$

$y = \frac{x^2}{8} - 2$

$\frac{dy}{dx} = \frac{2x}{8}$ at $x = 2$

$MT = 1/2$

$\therefore y + 3/2 = 1/2(x - 2)$

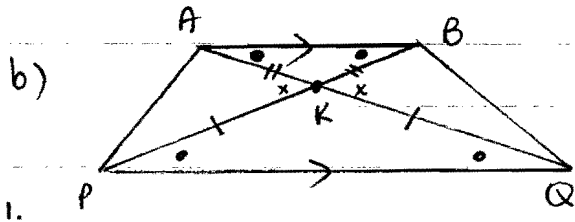
$2y + 3 = x - 2$

$2y = x - 5.$ ②



Question 4

a) $x^2 - 4x + y^2 + 6y = 12$
 $+4 \quad +9 \quad +13$
 $(x-2)^2 + (y+3)^2 = 25$
 $C = (2, -3) \quad r = 5$



i. $\angle KPQ = \angle KQP$ (equal angles opposite equal sides, $PK = QK$)

$\angle KBA = \angle KPQ$ (alt $\angle$'s $AB \parallel PQ$)

$\angle BAK = \angle KQP$ (alt $\angle$'s $AB \parallel PQ$)

$\therefore \triangle AKB$ is isosceles (2 equal $\angle$'s)

ii. In $\triangle AKP$ and $\triangle BKQ$

$PK = QK$ (given)

$AK = BK$ (equal sides opposite equal $\angle$'s, $\triangle AKB$ isosceles)

$\angle AKP = \angle BKQ$ (vertically opposite)

$\therefore \triangle AKP \equiv \triangle BKQ$ (SAS)

iii. $AP = BQ$, corresponding sides in congruent triangles

Question 5

a) let $m = 2^x \therefore m^2 - 9m + 8 = 0$

$(m-8)(m-1) = 0$

$m = 8$ or $m = 1$

i.e. $2^x = 8$ or $2^x = 1$

and $x = 3$ or $x = 0$

b) $3x^2 + 4 = Ax^2 + 4Ax + 4A + Bx + 2B + C$

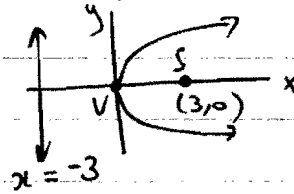
Equating coeff

$x^2 \quad 3 = A \quad A = 3$

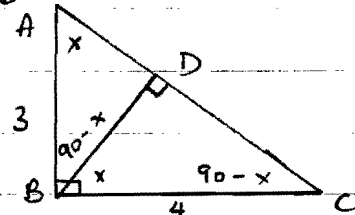
$x \quad 0 = 4A + B \quad B = -12$

$x^0 \quad 4 = 4A + 2B + C \quad C = 16$

c) Sideways $v = (0,0) \quad a = 3$



Question 6

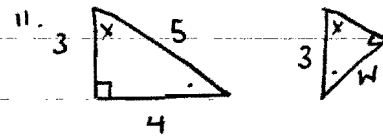


i. In $\triangle ABD$ and $\triangle ACB$

$\angle ADB = \angle ABC$ (given 90°)

$\angle A$ is common

$\therefore \triangle ABD \sim \triangle ACB$ (equiangular)



$\frac{w}{4} = \frac{3}{5}$ (ratio of corresp sides in $\sim \triangle$'s)

$\therefore 5w = 12$

$w = 2.4 \therefore BD = 2.4 \text{ cm}$

b) $3kx^2 - 5x + 3k = 0$

i. $\Delta = b^2 - 4ac$
 $= 25 - 4(3k)(3k)$

$\Delta = 25 - 36k^2$

ii. $3k < 0$ and $\Delta < 0$

$k < 0 \quad (5-6k)(5+6k) < 0$

$\therefore \text{Sol}^n: k < -5/6 \text{ only.}$

A i

Question 7

$$a) i. PA = \sqrt{(x+4)^2 + (y-0)^2}$$

$$PA^2 = (x+4)^2 + y^2$$

$$PB^2 = (x-4)^2 + (y-3)^2$$

$$ii. PA = 2PB$$

$$PA^2 = 4PB^2$$

$$x^2 + 8x + 16 + y^2 = 4[x^2 - 8x + 16 + y^2 - 6y + 9]$$

$$x^2 + 8x + 16 + y^2 = 4x^2 - 32x + 64 + 4y^2 - 24y + 36$$

$$3x^2 - 40x + 48 + 3y^2 - 24y + 36 = 0$$

$$3x^2 - 40x + 3y^2 - 24y + 84 = 0$$

b) let the roots be α and 2α

$$\alpha + 2\alpha = -b/a$$

$$\alpha \times 2\alpha = c/a$$

$$3\alpha = \frac{3k}{k-3}$$

$$2\alpha^2 = \frac{25}{k-3}$$

②

$$\alpha = \frac{k}{k-3} \quad \text{--- ①}$$

Sub ① into ②

$$2 \left[\frac{k}{k-3} \right]^2 = \frac{25}{k-3} \quad (k \neq 3)$$

$$2 \cdot \frac{k^2}{(k-3)^2} = \frac{25}{k-3} \quad \times (k-3)$$

$$\frac{2k^2}{k-3} = 25$$

$$2k^2 = 25(k-3)$$

$$2k^2 - 25k + 75 = 0$$

$$(2k-15)(k-5) = 0$$

$\therefore k = 15/2$ and $k = 5$.